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CreativityPrism: A Holistic Evaluation Framework for Large Language Model Creativity

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Abstract

Creativity is often seen as a hallmark of human intelligence. While large language models (LLMs) are increasingly perceived as generating creative text, there is still no *holistic* and *scalable* framework to evaluate their creativity across diverse scenarios. Existing methods of LLM creativity evaluation either heavily rely on humans, limiting speed and scalability, or are fragmented across different domains and different definitions of creativity. To address this gap, we propose CREATIVITYPRISM, an evaluation analysis framework that consolidates eight tasks from three domains, divergent thinking, creative writing, and logical reasoning, into a taxonomy of creativity that emphasizes three dimensions: quality, novelty, and diversity of LLM generations. The framework is designed to be scalable with reliable automatic evaluation judges that have been validated against human annotations. We evaluate 17 state-of-the-art (SoTA) proprietary and open-sourced LLMs on CREATIVITYPRISM and find that while proprietary LLMs dominate creative writing and logical reasoning tasks by a 15% lead over open-sourced ones, they offer no significant advantage in divergent thinking, a domain much less explored in existing post-training regimes. Our analysis also shows that high performance in one creative dimension or domain rarely generalizes to others; specifically, novelty metrics often show weak or negative correlations with other metrics. This fragmentation confirms that a holistic, multi-dimensional framework like CREATIVITYPRISM is essential for any meaningful assessment of LLM creativity [1].

1 Introduction

Creativity, the capacity to generate novel and valuable ideas or solutions [7, 19, 31], is a core human cognitive ability. It appears in many domains: crafting stories with surprising plot twists

[1] Project website: `https://joeyhou.github.io/CreativityPrism/`

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[4, 35], producing groundbreaking scientific discoveries [32, 63], solving problems under constraints [47, 71], or even expressing humor in everyday life [29, 76]. Its multifaceted nature has prompted extensive study in psychology and cognitive science, with efforts to capture creativity through both qualitative and quantitative approaches [2, 26, 52, 65].

Recently, with the rapid rise of general-purpose LLMs, interest has grown in probing their creativity [4, 10, 22, 47, 74]. But as with human creativity, machine creativity spans diverse and expansive contexts, making it difficult to define, formalize, and, above all, measure. Concretely, LLM creativity evaluation faces two challenges: **distinct definitions of creativity** across different domains and difficulty of **scalable, automatic evaluation** due to convoluted nature of creativity.

The first challenge stems from current research in machine creativity being scattered across different domains and focusing on narrow or singular dimensions. For example, the Divergent Association Task (DAT) [6, 12] and the Creative Short Story Task [35] emphasize lexical diversity; the Alternative Use Test (AUT) [22, 54] solely focuses on unconventional ideas of using daily items, overlooking the pragmatics of those solutions; CreativeMath [71] and NeoCoder [47] only study math and coding problems, correspondingly. What makes comparison even harder is that these task-specific and domain-specific benchmarks only benchmark their own choices of LLMs, which vary from one another. Without a comprehensive evaluation that incorporates the evaluation of creativity from all those dimensions, it is hard to uncover a holistic understanding of how well current state-of-the-art LLMs are doing when it comes to creativity.

The second challenge arises from the subjective nature of creativity, which makes it hard to automatically evaluate LLM output and leads many existing benchmarks to rely heavily on human evaluation [10, 69]. While human judgment is often considered the gold standard for nuanced tasks, it presents significant hurdles for modern AI research: it is prohibitively expensive, difficult to replicate at scale, and requires a long turnaround time that cannot keep pace with the current field. With new LLMs and model iterations being released nearly every week, a reliance on manual grading creates a massive bottleneck that prevents the rapid, iterative testing required for progress.

To this end, we propose CREATIVITYPRISM, a comprehensive and scalable evaluation evaluation framework of LLM creativity; it is holistic — consisting of eight tasks from three domains: divergent thinking, creative writing, and logical reasoning (i.e., mathematical reasoning and coding); it is also scalable — all evaluation metrics are automatic, ensuring easy benchmarking for any LLM. Specifically, we highlight following two strengths of CREATIVITYPRISM. First, comprehensiveness: we extend beyond a simple combination of existing benchmarks from various domains by systematically categorizing existing task-specific metrics along the three dimensions, i.e quality, novelty, and diversity, to facilitate a comprehensive measurement of model creativity (Figure 1). Every metric in CREATIVITYPRISM belongs to one of these three dimensions, and hence, evaluation results can be summarized into those three dimensions, providing a cross-domain and dimension-specific insight into LLMs' creativity performance. Second, scalability: we go above simple automatic evaluation with the LLM-as-a-Judge paradigm (§4.2). For every evaluation metric that requires LLM-as-a-Judge, we collect human judgments from well-trained researchers or domain experts, followed by Alternative Annotator Test [9] to verify our LLM-Judge setup can indeed achieve the same quality as well-trained human annotators or domain experts. In this way, we ensure the metrics in CREATIVITYPRISM are well aligned with high-quality human judgment.

We then evaluate 17 closed-sourced and open-sourced SoTA LLMs on CREATIVITYPRISM, to answer the following research questions. **RQ1: Where are the performance gap among LLMs' performance on creative tasks?** We found a notable performance gap between proprietary and open-sourced models, especially in logical reasoning tasks, followed by creative writing tasks. We also found the same performance gap to be most pronounced in the quality dimension. **RQ2: What are the correlations among LLMs' performance across different dimensions and domains?** In other words, would good performance in one creativity domain or dimension transfer to the other? Results have shown that the models perform similarly in metrics from the same task or the same domain; for metrics from different domains, models perform similarly in diversity and quality dimensions, while performances in novelty dimensions are much less correlated. We believe this stems from the inherent difference in how novelty is defined in different tasks and domains.

In short, our contribution can be summarized as follows:

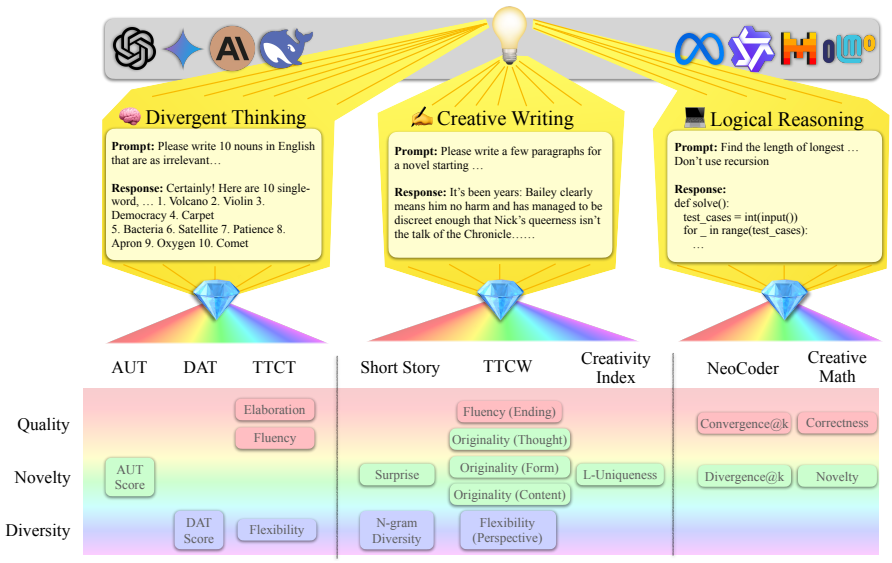

Figure 1: Overview of CREATIVITYPRISM. Each LLM is prompted to complete the tasks (example inputs in Table 1), and their outputs are evaluated using task-specific metrics. We also organize metrics into three dimensions of creativity: **quality**, **novelty**, and **diversity**.

- We select, combine, and refactor the codebase of eight creativity tasks across three domains to propose a easy-to-use, comprehensive and scalable evaluation framework, CREATIVITYPRISM;
- We propose the "quality, novelty, and diversity" taxonomy of creativity and categorize seventeen metrics into these dimensions for dimension level LLM creativity evaluation;
- We conduct high-quality human annotation to verify human agreement for each metric and human-LLM-Judge agreement during the automatic evaluation process;
- We benchmark 17 SoTA LLMs on CREATIVITYPRISM and conduct extensive analysis on model performance across domains and dimensions.

2 Related Work

Human Creativity The definition of creativity has varied across different domains. In psychology, Torrance Test of Creative Thinking (TTCT) [2] considers creativity as a combination of originality, flexibility, fluency, and elaboration; Runco & Jaeger [61] proposes a simpler taxonomy of originality and usefulness. In marketing, El-Murad & West [17], Rosengren et al. [60] considers advertisement creativity as the combination of usefulness and originality; additionally, Smith et al. [64] adds flexibility, fluency, elaboration, synthesis, and artistic values. In terms of creativity evaluation, Said-Metwaly et al. [62] summarizes more than 100 existing works and concludes that creativity evaluation is an "unsettled" issue, with one of the key reasons being the lack of holistic, cross-domain evaluation, which also motivates our work. A common belief among these work is the balanced view of quality (e.g., elaboration, usefulness), novelty (e.g., originality, synthesis), and diversity (e.g., flexibility) when it comes to different dimensions of creativity. This view directly inspire our taxonomy of creativity metrics in §3.

Machine Creativity Measurement of Machine creativity has become increasingly popular with the rapid development of LLMs. Many researchers attempt to provide a cross-domain or cross-dimension view of machine creativity in which a variety of angles have been taken, such as panoramic survey [34], computational models involved in machine creativity [21], the efficacy of evaluation methods and metrics [30, 44], the effect of LLM decoding strategy on output creativity [51], the tension between factuality and creativity [5], and multi-modal creativity [18, 70]. However, none of them attempt on validate, synthesize, and combine the evaluation metrics from a series of creativity tasks and benchmark existing SoTA LLMs as we do.